Università degli Studi di Padova

Dipartimento di Ingegneria dell'Informazione

Web Applications A.Y. 2025-2026
Homework 1 – Server-side Design and Development

Master Degree in Computer Engineering
Master Degree in Cybersecurity
Master Degree in ICT for Internet and Multimedia

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1 Objectives

The goal of this project is to design and develop a server-side Web application called **Lecture Notes App**: a platform that allows university students to upload, share, search, and download lecture notes in PDF format. Users can register and log in to the platform, organize notes by course, mark notes as favourites, and rate content shared by other students.

The main objectives of this project are:

- Provide a platform for sharing lecture notes
- Enable efficient search and organization of content
- Allow user interaction through ratings and favourites
- Ensure scalability and maintainability of the system

1.1 Technologies Used

The application is developed using:

- Java Servlets and JSP for server-side logic and dynamic content generation
- PostgreSQL as the relational database management system
- Apache Tomcat as the deployment server

1.2 Motivation

University students often struggle to find well-organized and reliable lecture notes. Existing solutions are either fragmented or lack quality control. This project aims to provide a centralized and user-driven platform to improve accessibility and collaboration.

2 Main Functionalities

This chapter describes the main functionalities offered by the application. The system is designed to manage user accounts, handle lecture notes, and provide interaction mechanisms between users.

2.1 User Management

The application allows users to create and manage their personal accounts. In particular, users can register, log in, log out, and update their profile information. These features are implemented through the following components:

- `RegisterServlet`
- `LoginServlet`
- `LogoutServlet`
- `ProfileServlet`

2.2 Note Management

Authenticated users can upload and manage lecture notes in PDF format. Each note is associated with a specific course and can be modified or removed by its owner. The system also allows users to download notes shared by others.

- `UploadNoteServlet`
- `UpdateNoteServlet`
- `DeleteNoteServlet`
- `DownloadNoteServlet`

2.3 Search and Organization

To facilitate access to content, the application provides a search functionality that allows users to find notes based on different criteria such as title, course, or author.

- `SearchNoteServlet`

Additionally, notes are organized by course, and a summary view is available to display all notes related to a specific course.

- `CourseSummaryServlet`

2.4 User Interaction Features

The platform includes features that enhance user interaction and content evaluation.

- **Favourites:** users can save notes in a personal favourites list and access them through a dedicated page (`favorites.jsp`).
- **Ratings:** users can rate notes uploaded by other students, contributing to an overall quality score.

3 Data Logic Layer

After presenting the main functionalities of the application, this chapter focuses on the data logic layer, which is responsible for modeling and managing the information handled by the system.

In particular, the structure of the database and the relationships between entities are described through an Entity-Relationship schema.

3.1 Entity-Relationship Schema

The database is designed to represent the core concepts of the application through a set of interconnected entities. The main entities are: `User`, `Note`, `Course`, `Favorite`, and `Rating`.

The relationships between these entities can be summarized as follows:

- A `User` can upload multiple `Notes`
- Each `Note` belongs to exactly one `Course`
- A `User` can mark multiple `Notes` as `Favorite`

- A `User` can assign a single `Rating` to a `Note`
- The `pair` (userId, noteId) is unique in the Rating entity.

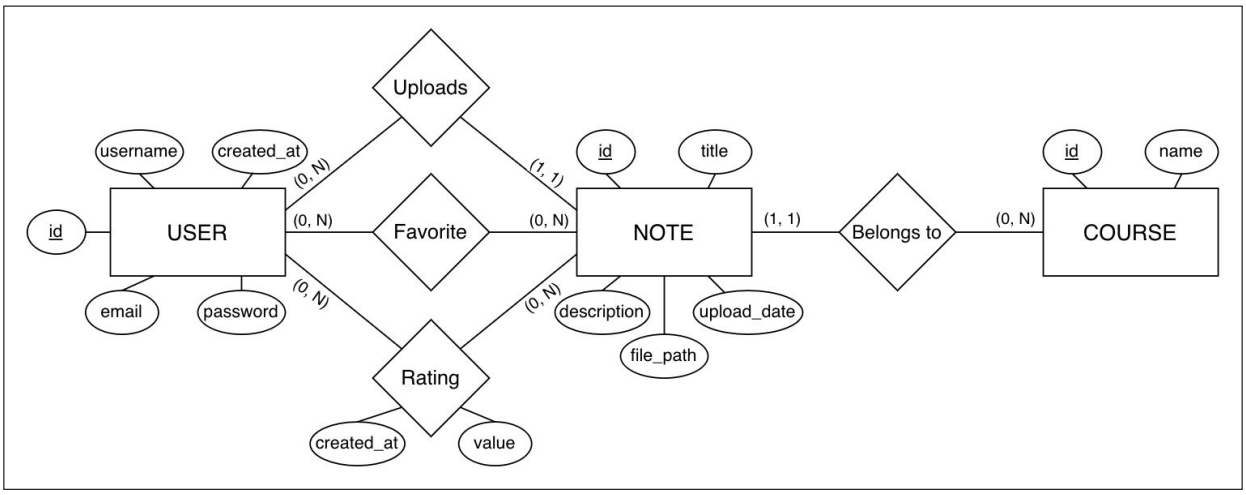

Figure 1: Entity-Relationship Schema of the Lecture Notes App

The ER diagram in Figure **??** provides a visual representation of the database structure and highlights the relationships between entities.

3.2 Entity Description

Each entity in the system stores specific information required to support the application functionalities.

Note. The `Note` entity contains the main information related to uploaded lecture notes, including: `id`, `authorId`, `authorUsername`, `courseId`, `courseName`, `title`, `description`, `uploadDate`, and `filePath`. The `filePath` attribute represents the location of the PDF file, which can be stored either on the server or on AWS S3 (as indicated by the `software.amazon.awssdk:s3` dependency in the project configuration).

User. The `User` entity stores authentication credentials and profile information required for user identification and interaction with the system.

Rating. The `Rating` entity represents the evaluation given by a user to a note. It links a `User` to a `Note` through a numeric score.

Favorite. The `Favorite` entity models the many-to-many relationship between users and notes, allowing each user to maintain a personalized list of favourite content.

Course. The `Course` entity groups notes by subject, enabling better organization and easier navigation within the application.

4 Presentation Logic Layer

After defining the data model of the application, this chapter describes the presentation layer, which is responsible for user interaction through the web interface.

The presentation layer is implemented using JSP pages located in `src/main/webapp/jsp/`. The application entry point is `login.jsp`, as specified in the `web.xml` configuration file.

4.1 Login Page – `login.jsp`

Presents a form for user authentication (username and password). On success the user is redirected to the home page.

4.2 Register Page – `register.jsp`

Allows new users to create an account by providing username, email, and password.

4.3 Home Page – `home.jsp`

The main dashboard after login. Displays the list of available notes with search and filter controls. Users can browse by course, download notes, or add them to favourites directly from this page.

4.4 Upload Page – `upload.jsp`

Form for uploading a new note: the user selects the course, inserts a title and description, and attaches a PDF file.

4.5 Profile Page – `profile.jsp`

Shows the user's personal information and the notes they have uploaded. Allows updating profile data and deleting own notes.

4.6 Favourites Page – `favorites.jsp`

Lists all notes that the logged-in user has marked as favourite.

Each page is designed to provide a simple and intuitive user experience, ensuring easy navigation and efficient interaction with the system.

5 Business Logic Layer

5.1 Class Diagram

The class diagram shown in Figure **??** provides an overview of the main components of the system and their relationships.

The architecture is structured into the following layers:

- **Controller Layer** – implemented through Java Servlets, responsible for handling HTTP requests and responses. Examples include `UploadNoteServlet`, `LoginServlet`, and `SearchNoteServlet`.
- **Model Layer** – composed of Plain Java Objects representing the core entities of the application, such as `User`, `Note`, `Course`, `Favorite`, and `Rating`.
- **DAO Layer** – responsible for data persistence and retrieval through JDBC interaction with the PostgreSQL database. Examples include `NoteDAO`, `UserDAO`, and `RatingDAO`.
- **Utility / Service Layer** – provides supporting functionalities such as database connection management and file storage. In particular, `DBConnection` handles database connectivity, while `StorageService` manages file upload and retrieval from cloud storage.

This layered design promotes loose coupling between components and simplifies future extensions of the system.

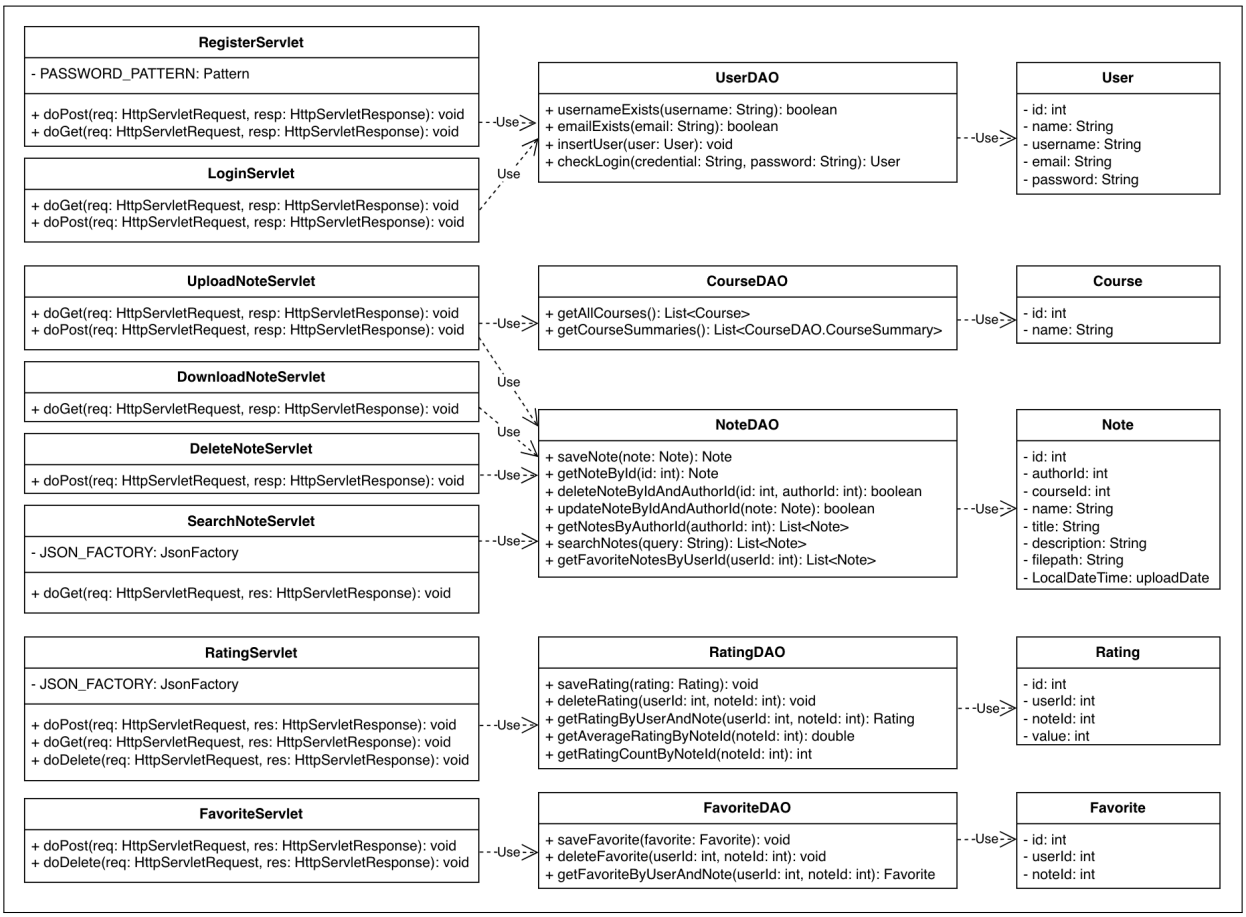

Figure 2: Class Diagram of the Lecture Notes App

5.2 Sequence Diagram

sequenceDiagram.pdf

5.3 REST API Summary

URI	Method	Description	Filter
What is the URI?	What HTTP method uses?	Describe briefly the result of calling the URI	is it behind a filter?

Table 2: Describe in this table your REST API

5.4 REST Error Codes

Error Code	HTTP Status Code	Description
Error Code Identifier	Corresponding HTTP Status Code	Description of the error

Table 3: Describe in this table your REST API

5.5 REST API Details

A resource

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- URL: `the URL to retrieve it`
- Method: `Method to retrieve it`
- URL Parameters:
- Data Parameters:
- Success Response:
- Error Response:

6 Group Members Contribution

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